

Aakarsh Sangar

Research CV

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Engineering Portfolio: aakarshsangar.com

EDUCATION

Purdue University

West Lafayette, IN | Aug 2026 - Present

First-Year Engineering; Intended B.S. Mechanical Engineering

Canyon Crest Academy

San Diego, CA | Diploma, 2026

17 APs including Calculus BC, Physics C: Mechanics, Physics C: E&M, Physics 1/2, Statistics, Computer Science Principles

RESEARCH INTERESTS

Focus: Robotics, mechatronics, robotic manipulation, computer vision, and simulation for hardware systems that connect mechanical design with perception-informed control.

INTERNSHIPS

ModalAI

San Diego, CA | Sep 2024 - Jan 2025

Drone Hardware & Computer Vision Intern

- One of three interns on a pre-release Stinger VisionM FPV kit spanning hardware bring-up, companion-computer software, and customer documentation.
- Integrated and configured VOXL SDK 1.4.5; stood up onboard computer vision for limited classroom object recognition.
- Wrote the kit instruction manual for assembly and first bring-up, translating early hardware/software setup into repeatable user documentation.

RESEARCH AND TECHNICAL PROJECTS

Planetary-Drive Robotic Arm

Jul 2026

Onshape, ROS 2

- Created a robotic arm around a custom planetary-drive actuator concept sized to the torque envelope of the arm joints.
- Designed independently actuated base, link, wrist, and gripper joints to evaluate arm structure, axis definitions, and motion interfaces.
- Actively learning ROS 2 and running the arm development environment through a VNC server on my Mac.

Actuator Study & Simulated Arm

Sep 2025

UCSD faculty-guided | Onshape, URDF, PyBullet

- Compared servo, stepper, BLDC, pneumatic/hydraulic, and series-elastic actuators across torque density, controllability, power, and integration.
- Exported an articulated Onshape arm to a URDF tree and validated joints, axes, limits, and mesh frames in PyBullet.

FIRST Tech Challenge 18263 HIGH VOLTAGE

San Diego, CA | 2020 - May 2026

CAD Lead, Build Team

- Led Onshape CAD and mechanical iteration on a multidirectional competition robot chassis with PID stabilization.
- Integrated AprilTag localization and kinematics-based turret aiming; awards include 2x Design, Motivate, Innovate, and Inspire 3rd Place.

rPPG Pulse Pipeline

Jun 2026

Cal Hacks AI (Berkeley)

- Built a video-to-pulse pipeline with face ROI tracking, color-channel preprocessing, temporal filtering, and pulse extraction.
- Designed the signal workflow around motion and illumination noise in webcam-based measurements.

Faultline

2026

CalHacks AI | Python, LLM APIs, AI agents, information retrieval

- Developed a multi-agent research assistant to analyze academic literature and identify conflicting claims between papers.
- Built LLM workflows to retrieve papers, summarize findings, compare methodologies, and propose validation experiments.

TEACHING

Canyon Crest Academy

San Diego, CA | Aug 2025 - Dec 2025

Teaching Assistant, Advanced Engineering

- Mentored students in Onshape, circuit design, Arduino, and structured engineering problem-solving.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Arduino

Robotics & CV: URDF, PyBullet, VOXL SDK, PID, AprilTag, onboard object recognition

CAD & Hardware: Onshape Certified, actuator/transmission selection, planetary drives, circuit design

Signals & AI: rPPG, temporal filtering, multimodal vision APIs, LLM APIs, AI agents, information retrieval